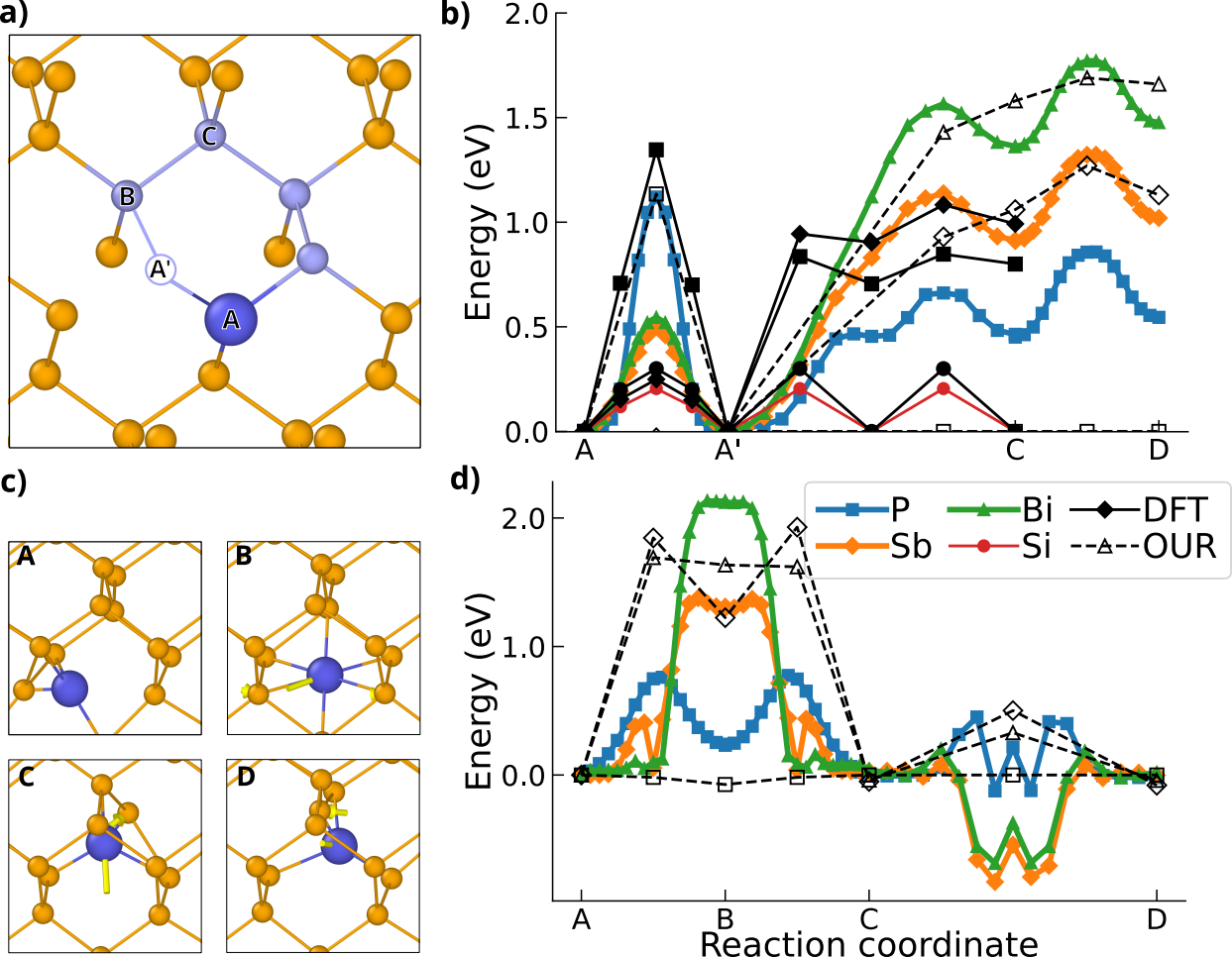

a)
C
B
A'
A
b)
2.0
1.5
1.0
0.5
0.0
Energy (eV)
A
A'
C
D
c)
A
B
C
D
d)
P
Bi
DFT
Sb
Si
OUR
2.0
1.0
0.0
Energy (eV)
A
B
C
D
Reaction coordinate